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Problem Chosen:	B

High-Speed Train Aerodynamic Shape Optimization Research

Abstract

With the rapid development of the computing field, the pursuit of high speed and low noise pollution of the aerodynamic shape is particularly important. In this paper, a corresponding mathematical model has been established to study the speed as well as noise domain of high speed railways and simulations have been carried out.

For problem 1, based on Standard $k - \varepsilon$ Models, four preliminary high-speed train models are established respectively, and the force analysis of the universal train head is carried out in two dimensions, as well as the simulation of the four models by RWIND wind tunnel software, and the Δ factors used for description are defined, and the Δ factors of each model are compared, and it is concluded that **TP1 is the best high-speed train shape with the lowest air resistance.**

For problem 2, we divided TP1 into 5 parts, optimized and fine-tuned them by Pareto search method, and built the optimized high speed rail model.

In response to problem 3, and the fact that many scholars have studied the noise pollution of high-speed trains, many countries have also introduced corresponding speed limit regulations. We model and analyze a high-speed train traveling at 85m/s based on the physical properties of turbulence, and conclude that **TP4 is the optimal high-speed train shape that generates the least noise.**

Finally, we combine the characteristics of TP1 and TP4 and use a multi-objective particle swarm algorithm to design a new train which has a more balanced force distribution, i.e., good upper speed limit, and low noise to the environment in wind tunnel experiments, and the Δ of the new train is excellent when compared by the Δ factor. $\Delta_{TP1} = 0.0105$, $\Delta_{TP4} = 0.0031$, $\Delta_{TP_{best}} = 0.0029$. Comparing the models before and after optimization, it can be found that the nose cone height decreases, the nose cone length increases, and the nose cone becomes smoother; the driver's chamber height is adjusted downward, and the front half of the streamline shape becomes narrower and the width of the rear half increases.

Keywords: High-Speed Train Standard $k - \varepsilon$ Models Wind Tunnel Simulation MOPSO Blender Modeling Software

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I. Introduction

1.1 Problem Background

In order to indicate the origin of problems, the following background is worth mentioning. High-speed high-speed rails have become the way of travel for most people, and the continuous research on high-speed high-speed rails makes the development of high-speed high-speed rails play a role in promoting, and with the accelerated pace of people's lives and improve the transport capacity and transport efficiency of high-speed railroads, railroad speed has become the current development trend, which needs to be continuously optimized, and this paper analyzes the air resistance to its high-speed railroad head, generates noise and other problems, and establishes an optimization model.

1.2 Restatement of the Problem

Question 1

Establish the mathematical model of air resistance of high speed railroad, analyze the force relationship of high speed railroad head in general weather and extreme weather, and simulate accordingly out the force relationship of cone type and four kinds of high-speed railroad heads in the figure. The best high-speed railroad shape with the lowest air resistance is obtained.

Question 2

Analyze the effect of the curvature of the head of the high speed rail on the air resistance, and in order to minimize the air resistance, build an optimization model, design the best shape of the high speed rail, and draw a sketch of the optimized high speed rail.

Question 3

The noise intensity generated by the conical shape as well as the four classical high speed high-speed rails shown in Figure 2 is analyzed, and the noise distribution is simulated to select the best shape of high speed high-speed train with the least noise.

Question 4

The results of the above three problems were combined to create an optimized model of the shape of the high speed rail, to design the best shape of the high speed rail to increase its speed and reduce its noise, and to draw a sketch of its shape and its structural parameters.

1.3 Our Work

Question 1

Through our analysis of the continuity principle of the fluid, in essence, the fluid in the flow of mass conservation, after Bernoulli's equation can be obtained for the ideal fluid it is the conservation of mechanical energy, through the Poiseuille theorem can be obtained for the actual fluid is the existence of viscous friction, this viscous friction will cause a kind of resistance to the relative motion of fluid and solid, according to this establishment of air resistance model to study high-speed railroad Air resistance. When the fluid speed is very fast, turbulence will be generated. In this regard, the standard $k - \varepsilon$ model is applied to solve the turbulent kinetic energy and its dissipation rate equation. For the four high-speed railroad shapes, in extreme weather (such as rain and snow), we use the force analysis method to represent the force relationship, for the different shapes of high-speed railway we make four models of high-speed railway front end by Blender software, and the four models are carried out wind tunnel experiments, which can intuitively feel the air resistance to each part, and by comparing the air resistance distribution to select the By comparing the air resistance distribution, we choose the shape of high-speed railroad with the least air resistance. **The final TP1 is the best high speed rail model with the least air resistance.**

Question 2

On the basis of TP1, an optimization model is established based on the Pareto search process to design the optimal high speed rail model with the minimum air resistance.

Question 3

We firstly collect the relevant noise data to have a certain understanding of the noise of high-speed train (train) in each country, and secondly get that when the speed of the car is very fast, the wave surface accelerates the buildup, which makes the air friction increase and the noise also increases. In this regard, the external aerodynamic noise model of high-speed train is established, and the turbulent kinetic energy turbulent dissipation rate at each node is extracted by the air resistance model in turbulence, so as to determine the sound power at each node. Then the *Lighthill-Curle acoustic analogy theory* is compared to the high-speed railroad pressure distribution to realize the aerodynamic noise simulation. **TP4 is finally determined to be the optimal shape of the high-speed railway that generates the least noise.**

Question 4

Based on the Pareto search in the second question, we used the multi-objective particle swarm algorithm (MOPSO) to combine the characteristics of TP1 and TP4 to find the approximate model, and the model was simulated in a wind tunnel to obtain the relevant data.

Combined with the simulated values of the theoretical conic train in the third question, the optimized model $\Delta_{TP_{best}} = 0.0029$ is the smallest among the three of TP1 and TP4 in terms of Δ coefficients (the less the better), reducing **72.38%** compared to TP1 and reducing **6.45%**, which verifies the feasibility of the model.

II. Model Assumptions

1. The material is the same for different high-speed railroads.
2. High-speed rail is straight.
3. The wind speed is the same.
4. Body length is the same.

III. Symbol Description

Symbols	Meaning	Unit
u	Turbulence velocity	m/s
c_0	Air speed	m/s
S	Cross-sectional area	m^2
p_1	Pressure	Pa
R	Radius	m
P_{ij}	Pulsating pressure of air on the outer surface of the body	Pa
P	Sound pressure	Pa
P_0	Reference sound pressure	Pa

IV. Models

4.1 Physical basis of the model

4.1.1 Principle of Continuity

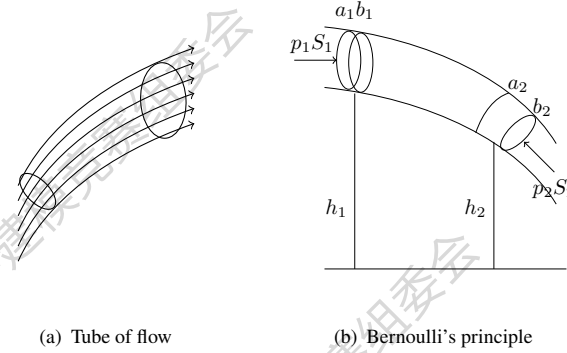


Figure 1 The basic theory of ideal fluids.

In a constant flow field, take any section of the flow tube and set the vertical cross sections at its two ends as dS_1 and dS_2 . In a constant flow, the tube of flow is stationary and the density ρ at each point in the fluid should not change with time, so the mass of the fluid in this section is constant, and thus the mass flow rate dQ_{m1} from one section is always equal to the mass flow rate dQ_{m2} from the other end. So

$$\rho_1 v_1 dS_1 = \rho_2 v_2 dS_2 \quad (1)$$

Or, along the arbitrary tube of flow

$$\rho v dS = \text{Constant} \quad (2)$$

If we further assume that the fluid is incompressible, then its density does not change and we have $\rho_1 = \rho_2$, thus

$$v_1 dS_1 = v_2 dS_2 \quad (3)$$

Or, along the arbitrary tube of flow

$$v dS = \text{Constant} \quad (4)$$

The above equations are called the continuity principle of fluids, which in physical essence embodies the conservation of mass in the flow of fluids.

4.1.2 Daniel Bernoulli's Equation

As shown in Figure 1, a section of fluid is sectioned off by cross sections S_1 and S_2 for any of the flow tubes in an ideal fluid with constant flow. During the time interval Δt , S_1 at the left end moves from position a_1 to b_1 and S_2 at the right end moves from position a_2 to b_2 , such that $\overline{a_1 b_1} = \Delta l_1$ and $\overline{a_2 b_2} = \Delta l_2$, then $\Delta V_1 = S_1 \Delta l_1$ and $\Delta V_2 = S_2 \Delta l_2$ are the volumes of fluid flowing in and out during the same time interval, respectively, for the incompressible fluid constant flow, $\Delta V_1 \equiv \Delta V_2 \equiv \Delta V$. Because there is no viscosity, i.e., no dissipation, we can apply the law of

conservation of mechanical energy to the fluid in this section of the flow tube. Although the fluid changes in the section from b_1 to a_2 , the flow is constant and the state of motion remains unchanged. Thus, the kinetic and potential energies remain unchanged. Therefore, when examining the change in energy, we only need to calculate the difference in capacity between the two end elements ΔV_2 and ΔV_1 . First, consider the change in kinetic energy:

$$\Delta E_k = \frac{1}{2} \rho v_2^2 \Delta V - \frac{1}{2} \rho v_1^2 \Delta V$$

Consider again the change in gravitational potential energy:

$$\Delta E_p = \rho g (h_2 - h_1) \Delta V$$

Now look at the work done by the external force on the fluid in this section of the flow tube. Let the pressure at the left end be p_1 and the force $F_1 = p_1 S_1$ acting on S_1 . The work done by the external force is $A_1 = F_1 \Delta l_1 = p_1 S_1 \Delta l_1 = p_1 \Delta V$; the pressure at the right end is p_2 , the force acting on S_2 The force on $F_2 = p_2 S_2$, the external work is $A_2 = -F_2 \Delta l_2 = -p_2 S_2 \Delta l_2 = -p_2 \Delta V$. Therefore

$$A_{out} = A_1 + A_2 = (p_1 - p_2) \Delta V$$

From the conservation of mechanical energy $A_{out} = \Delta E_k + \Delta E_p$ we get

$$(p_1 - p_2) \Delta V = \frac{1}{2} \rho (v_2^2 - v_1^2) \Delta V + \rho g (h_2 - h_1) \Delta V$$

So

$$p_1 + \frac{1}{2} \rho v_1^2 + \rho g h_1 = p_2 + \frac{1}{2} \rho v_2^2 + \rho g h_2 \quad (5)$$

Since 1 and 2 are any two points in the same stream, Equation 5 can also be expressed as the same *streamline*.

$$p + \frac{1}{2} \rho v^2 + \rho g h = \text{Constant} \quad (6)$$

Equation 5 and 6 are the *Daniel Bernoulli's Equation*.

4.1.3 J. L. M. Poiseuille's Equation

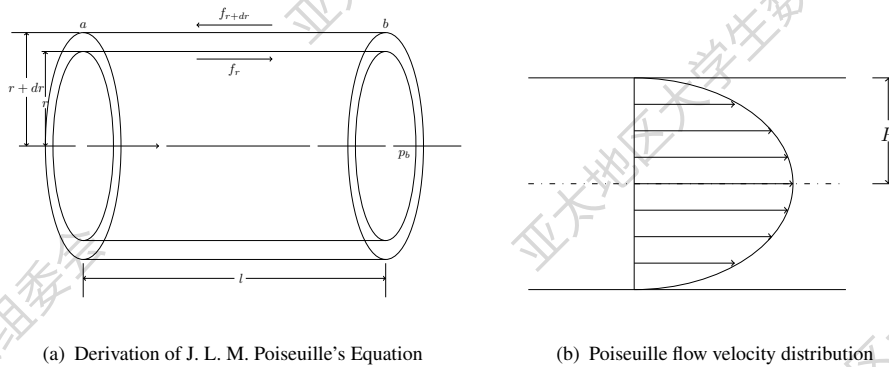


Figure 2 Derivation of J. L. M. Poiseuille's Equation and Flow Rate Distribution.

Due to the existence of viscosity, attached to the wall of the solid immersed in the fluid and the relative speed of the solid surface is always 0. To keep the fluid for constant flow, there must be

pressure; to keep the solid for uniform motion, there must be a support force. This means that the relative motion of the fluid and the solid is subject to a resistance. This resistance comes from the viscous nature of the fluid^[2].

Consider a horizontal tube ab of radius R and length l , in which the fluid flows along the axis and the velocity of the fluid attached to the wall of the tube is 0 due to viscosity. For a given pressure difference, the velocity of the fluid has a distribution along the radial direction, with the maximum velocity v in the central tube axis ($r = 0$), decreasing to 0 around $r \rightarrow R$. In the following we first determine the radial distribution function $v(r)$ of the velocity, for which it is envisaged to isolate a cylindrical thin flow layer within the fluid with inner and outer radii r and $r + dr$ respectively, with side areas of $2\pi rl$ and $2\pi(r + dr)l$ respectively, subject to viscous forces ¹ respectively.

$$\begin{cases} f_r = -\eta \left(\frac{dv}{dr} \right)_r 2\pi rl \\ f_{r+dr} = \eta \left(\frac{dv}{dr} \right)_{r+dr} 2\pi(r + dr)l \end{cases}$$

Here the velocity gradient $\frac{dv}{dr} < 0$, the positive and negative signs in Eq. are determined by specifically analyzing the direction of the viscous forces on both sides of this thin layer. The combined force of the viscous forces on the flow layer is:

$$f = f_{r+dr} + f_r = 2\pi\eta l \left[(r + dr) \left(\frac{dv}{dr} \right)_{r+dr} - r \left(\frac{dv}{dr} \right)_r \right] = 2\pi\eta l \frac{d}{dr} \left[r \left(\frac{dv}{dr} \right) \right] dr$$

In the case of constant flow this force should be balanced by the pressure difference on the end face $2\pi r dr$.

$$(p_b - p_a) 2\pi r dr = 2\pi\eta l \frac{d}{dr} \left(r \frac{dv}{dr} \right) dr \implies d \left(r \frac{dv}{dr} \right) = \frac{p_b - p_a}{\eta l} r dr$$

The two sides are integrated from $r = 0$ to r , later then integrate from r to R , we get:

$$r \frac{dv}{dr} = \frac{p_b - p_a}{2\eta l} r^2, \text{ or } dv = \frac{p_b - p_a}{2\eta l} r dr$$

$$v(R) - v(r) = \left[\frac{p_b - p_a}{4\eta l} r^2 \right]_r^R = \frac{p_b - p_a}{4\eta l} (R^2 - r^2)$$

Because $v(R) = 0$ on the wall of the tube, the final radial distribution of the flow velocity in the tube is obtained as:

$$v(r) = \frac{p_a - p_b}{4\eta l} (R^2 - r^2) \quad (7)$$

It has the form of a rotating paraboloid. Now calculate the flow rate, the flow rate through the circular area $2\pi r dr$ is $dQ_v = 2\pi v r dr$, so the total flow rate in the tube is:

$$Q_v = 2\pi \int_0^R v(r) r dr = \frac{\pi(p_a - p_b)}{2\eta l} \int_0^R (R^2 - r^2) r dr = \frac{\pi}{8} \frac{p_a - p_b}{\eta l} R^4 \quad (8)$$

The average velocity $\bar{v}$ in the tube can be defined as the volume flow Q_v divided by the cross-sectional area $S = \pi R^2$: $\bar{v} = \frac{Q_v}{\pi R^2}$. In addition, the pressure difference $p_a - p_b$ divided by the tube length l is the pressure gradient along the tube, and multiplied by the cross-sectional area of the tube S is the pressure gradient, expressed by $\frac{\Delta F}{\Delta l}$, so that by equation 8 we have:

$$\frac{\Delta F}{\Delta l} = \frac{p_a - p_b}{l} \pi R^2 = \frac{8\eta Q_v}{R^2} = 8\pi\eta \bar{v} \quad (9)$$

¹ $P = 1 \text{ dyn} \cdot \text{s/cm}^2 = 0.1 \text{ N} \cdot \text{s/m}^2 = 0.1 \text{ Pa} \cdot \text{s}$

4.2 Air Resistance model and Force Analysis of the high-speed train

4.2.1 Air Resistance model

Turbulent Flow and Reynolds Count

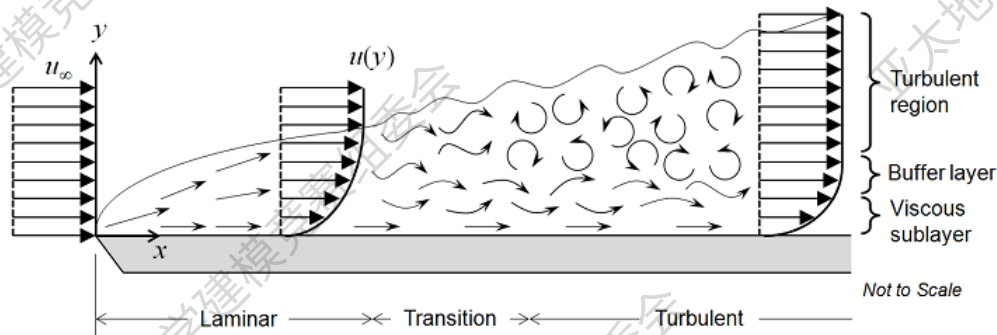


Figure 3 Turbulent flow schematic.

Turbulence we can think of as the motion of a superposition of various vortices. There are many models of turbulent flow, but they can be broadly grouped into the following three categories:

1. The first type is the *turbulent transport coefficient model*, proposed by Boussinesq in 1877 for two-dimensional flows, which expresses the second-order correlation of the velocity pulsation as the product of the average velocity gradient and the turbulent viscosity coefficient.
2. The second type is to discard the concept of turbulent transport coefficients and directly establish *transport equations* for turbulent stresses and other second-order correlated quantities.
3. The third category is *large turbulent simulations*. The first two types are based on the statistical structure of turbulence and statistical averaging of all vortices. The large turbulent simulations divides turbulence into large scale turbulence and small scale turbulence and obtains the characteristics of the motion of large eddies by solving the three-dimensional modified *Navier-Stokes Equation*, while the above approach is still adopted for the motion of small eddies.

Figure 3 is a diagram of the vortex flow². In practice, the model chosen for the solution is determined by the characteristics of the specific problem. The general principles of selection are high accuracy, simplicity of application, saving time of calculation, and generality at the same time.

In this paper, the viscosity of fluid is considered and the *standard $k - \varepsilon$ model*, which is the most widely used, stable and relatively accurate in engineering applications, is used to simulate the high speed rail and avoid the limitations of wall calculation.

Standard $k - \varepsilon$ Models

The the standard $k - \varepsilon$ model requires the solution of the turbulent kinetic energy and its dissipation rate equations. The turbulent kinetic energy transport equations are obtained by exact equation derivation, but the dissipation rate equations are obtained by physical derivation and mathematically simulating similar proto-equations. The model assumes that the flow is fully turbulent and the effect of molecular viscosity can be neglected. Therefore, the standard $k - \varepsilon$ model is only suitable for the

²Image from <https://zhuanlan.zhihu.com/p/21338549>

simulation of fully turbulent flow processes. The turbulent kinetic energy k and dissipation rate ε equations of the standard $k - \varepsilon$ model have the following form.

$$\frac{\partial}{\partial t}(\rho k) + \frac{\partial}{\partial x_i}(\rho k u_i) = \frac{\partial}{\partial x_j} \left[\left(\mu + \frac{\mu_t}{\sigma_k} \frac{\partial k}{\partial x_j} \right) \right] + G_k + G_b - \rho \varepsilon - Y_M + S_k \quad (10)$$

$$\frac{\partial}{\partial t}(\rho \varepsilon) + \frac{\partial}{\partial x_i}(\rho \varepsilon u_i) = \frac{\partial}{\partial x_j} \left[\left(\mu + \frac{\mu_t}{\sigma_\varepsilon} \right) \frac{\partial \varepsilon}{\partial x_j} \right] + C_{1\varepsilon} \frac{\varepsilon}{k} (G_k + C_{3\varepsilon} G_b) - C_{2\varepsilon} \rho \frac{\varepsilon^2}{k} + S_\varepsilon \quad (11)$$

where G_k denotes the turbulent kinetic energy generation due to the mean velocity gradient; G_b denotes the turbulent kinetic energy generated by buoyancy; Y_M denotes the contribution of fluctuating expansions in compressible turbulence to the total dissipation rate; S_k and S_ε are user-defined source terms; $C_{i\varepsilon}$ ($i = 1, 2, 3$) denotes the constants; σ_k and σ_ε denote the turbulent Prandtl number of turbulent kinetic energy k and dissipation rate ε .

4.2.2 Force Analysis

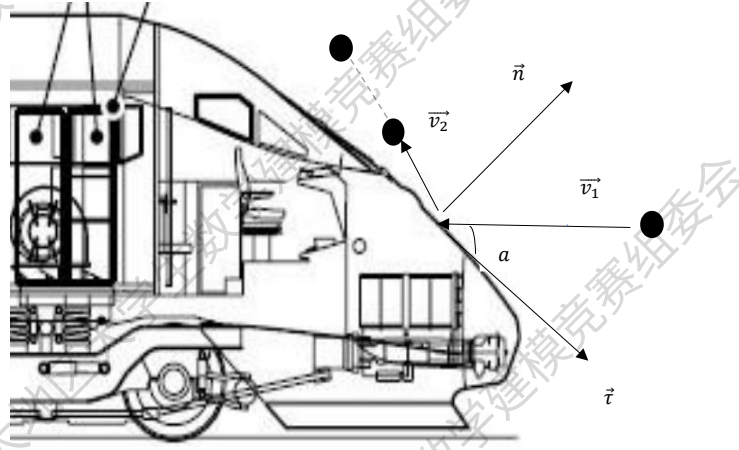


Figure 4 Particle-based force analysis.

Considering the extreme weather, such as rain, snow and wind conditions, because of the fast speed of the train relative to the rain and snow, so the rain and snow will also impact the head of the train at a very fast speed. Figure 4 shows the trajectory of the rain and snow when it touches the surface of the head of the train, when the particle touches the wall, the tangential force component by $\vec{\tau}$ has a small value, and the surface normal direction $\vec{n}$ on the z axis provides the particle motion. When the collision nature is fully elastic, $|v_1| = |v_2|$ motion rate is the rate in front of the contact wall, while in the actual impact is inelastic, that is, $|v_1| > |v_2|$, so if you want to reduce the impact of rain and snow on the head of the train, you can choose the elastic material coating.

4.2.3 Simulation of Conical and Four Typical models

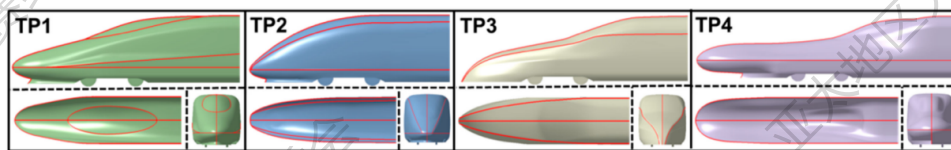


Figure 5 Simplified models of four typical high-speed train head structures^[5].

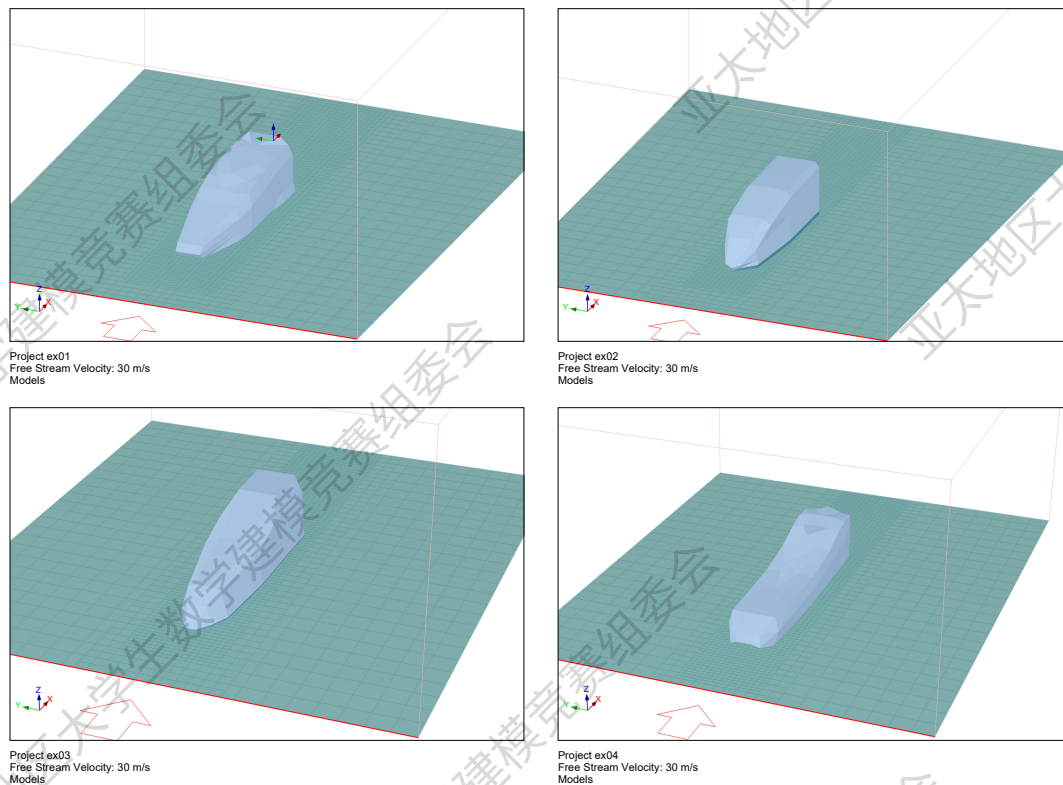
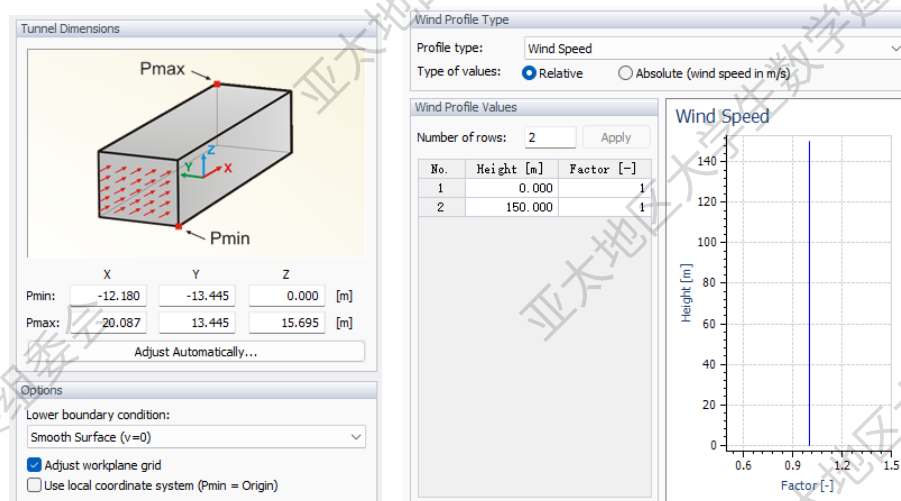


Figure 6 Use Blender software to create the models and then put them into RWIND.

We first create what we think the head of the high speed train will look like by using *Blender* software in conjunction with *Figure 5* provided in the topic, as shown in *Figure 6*. We first consider the general case that comes with the software to verify the feasibility of the four head models. After setting the corresponding parameters, we conduct wind tunnel experiments on each of the four models, as shown in *Figure 8*, it can be found that the vortex and the *Phenomenon of the Poiseuille* leaf appear at the rear of the front end, we continue to simulate the surface pressure of the front end and get as shown in *Figure 9*.



(a) Setting of simulation parameters

(b) Setting of simulation parameters

Figure 7 Setup of RWIND.

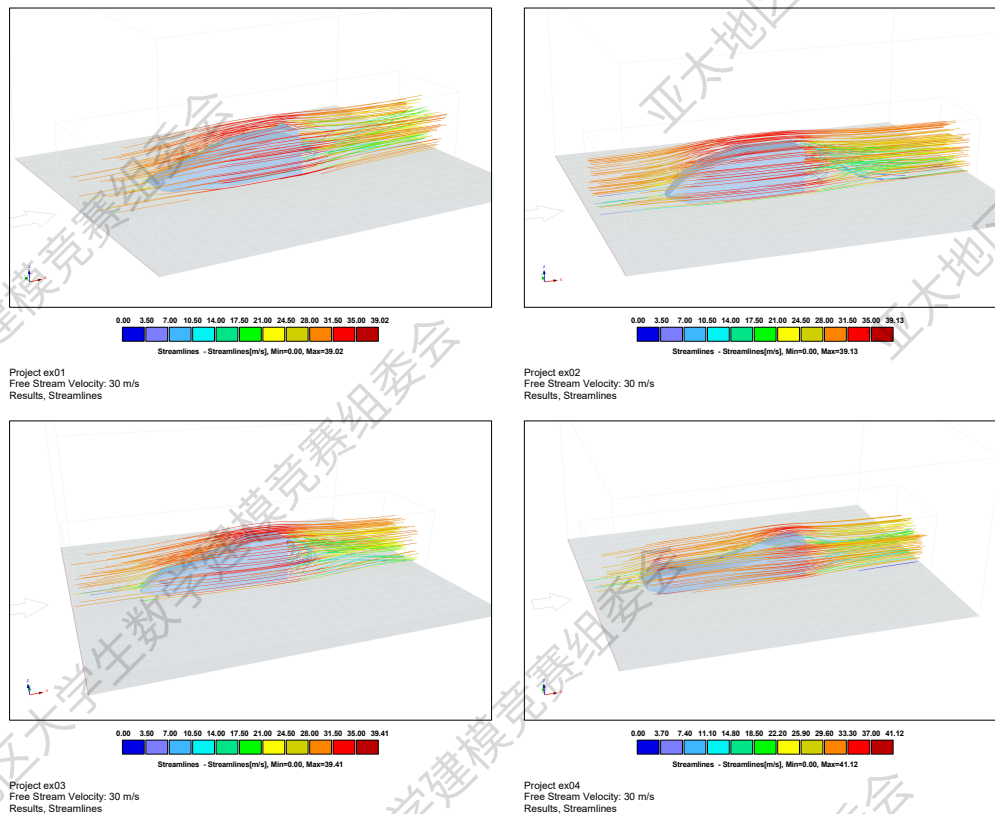


Figure 8 Simulate the streamline of the front surface with RWIND software.

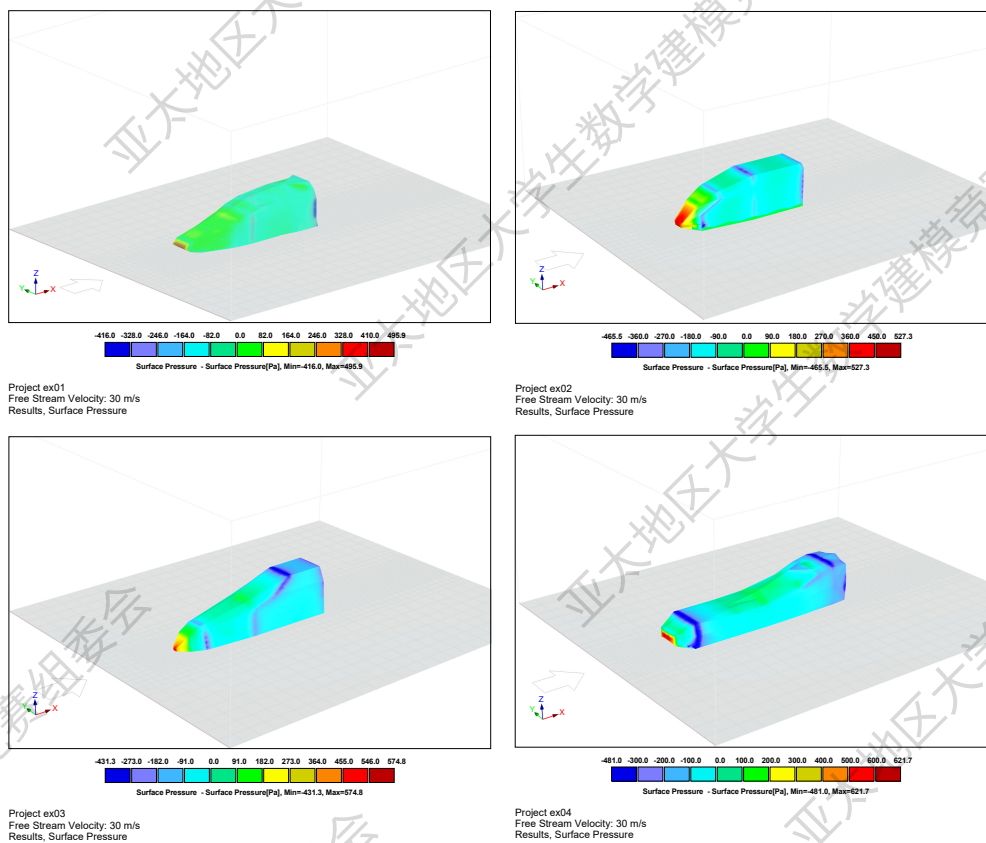


Figure 9 Pressure simulation of the head of a high speed train.

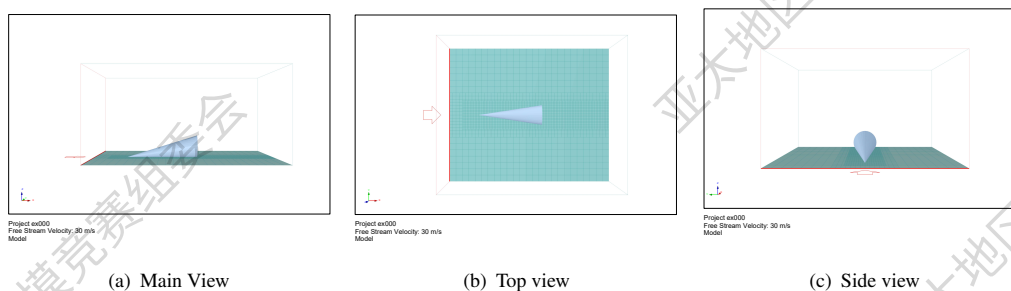


Figure 10 Theoretical conical type.

As shown in *Figure 10*, we designed a theoretical head model according to the requirements of the topic, and used it to compare with other four types of typical high-speed trains. In combination with the *Figure 9*, we set up a set of TP1 and theoretical conical type for comparison.

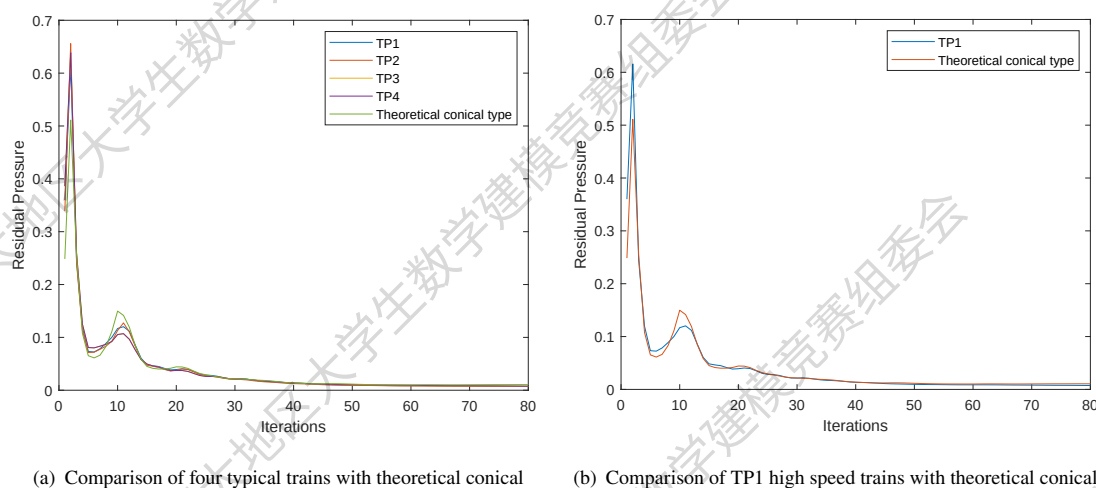


Figure 11 Comparison of high speed rail with theoretical conical.

The data of relevant simulations are exported from *RWIND* software, and we use the following formula to calculate the difference between four typical high speed rail $x_l^{(TP_i)}$ and the theoretical values x^* , and the results are presented in Table 1, where N equals 300.

$$\Delta = \sum_{j=1}^N \left(x_j^{(TP_i)} - x^* \right)^2 \quad (12)$$

Table 1 The difference between four typical high speed rail and theoretical values.

	TP1	TP2	TP3	TP4
Δ	0.0695	0.1031	0.0885	0.1120

From the difference Δ : Among these four typical HSRs, **TP1 is the best HSR shape with the lowest air resistance**. Detailed values can be found in *Table 6*.

4.3 Optimization model of high-speed train head

4.3.1 Introduction to Pareto Optimality

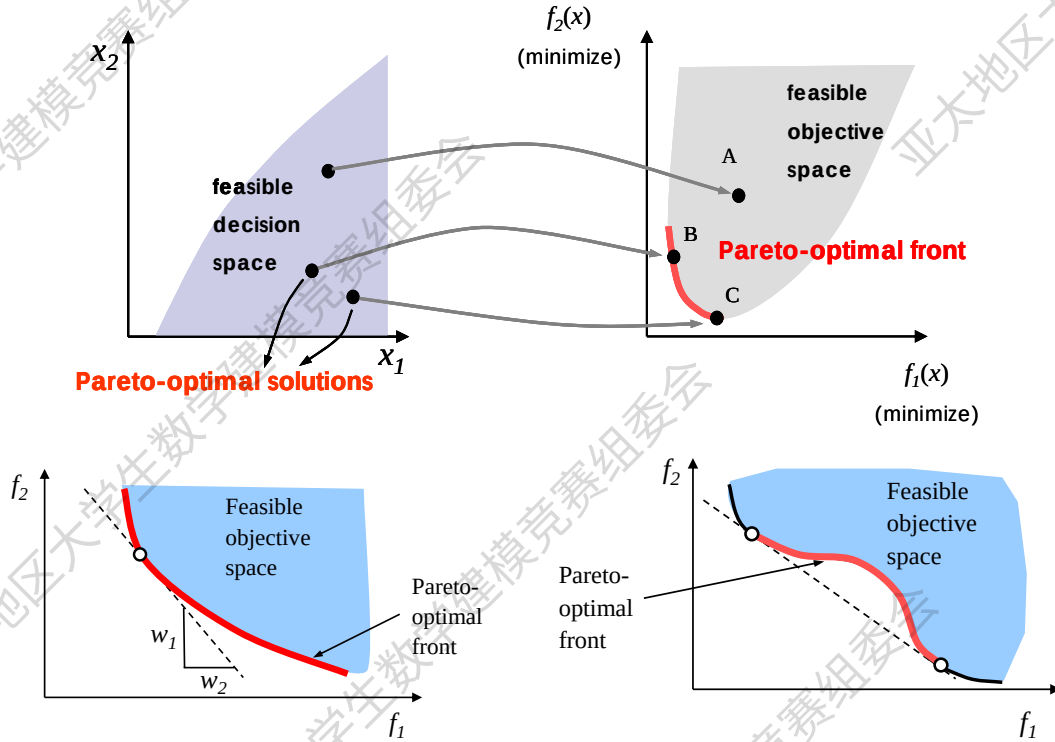


Figure 12 Pareto optimality schematic.

The deformation of each region is done sequentially, and after obtaining the coordinate increments of all the deformed regions, the deformed surface is obtained by adding the coordinate increments with the original coordinate values. In the optimization analysis, a total of 5 optimization variables were designed, which are Figure 13 Zone1-5 (Because Zone6 belongs to the carriage part, this paper does not consider for the time being).

There are usually several methods for solving multi-objective planning as follows.

1. *multi-objective into single-objective*, synthesizing multiple objectives into a single objective function weighted by their importance, or using the ideal point method, i.e., first finding each single-objective optimal function value and then expressing the objective function as the Euclidean distance to these single-objective optima.
2. *stratified sequence method*, a sequence is given by the importance of the objectives, and each time the optimal solution of the next objective is found within the optimal solution set of the previous objective until a common optimal solution is found.
3. *Pareto optimality search*, as shown in Figure 12 that is, the set of *Pareto solutions* is found ³, for Pareto solutions, the improvement of one goal is at the expense of the deterioration of the other goal.

³Image from <https://hpzhao.github.io>

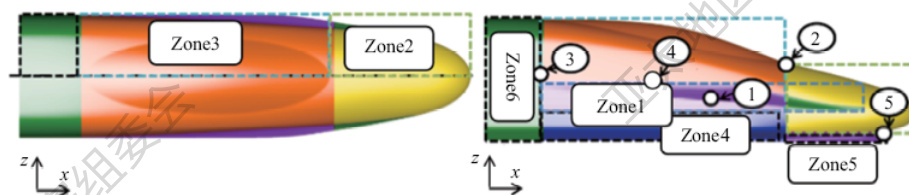


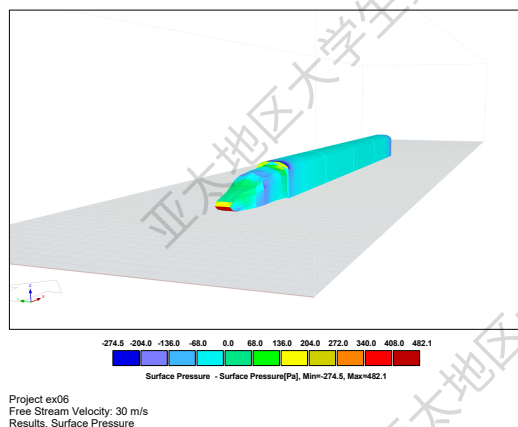
Figure 13 Different areas of streamlined high-speed train head structure^[6].

4.3.2 Optimal numerical solution

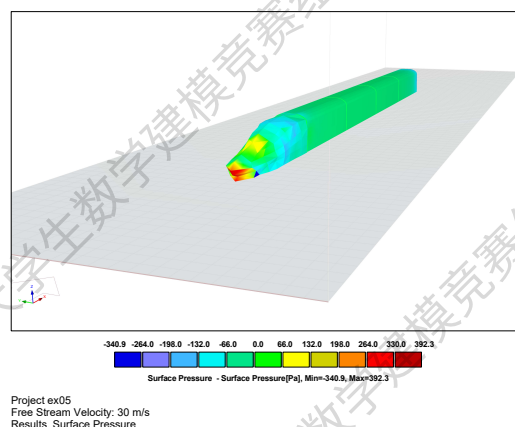
The introduction of weighting factors to weight multiple objectives is transformed into a single-objective planning problem to be solved in a model with parameters, which is actually a series of models that vary with reference. Solving a series of models is achieved using the *for loop* and a series of useful results are saved. The following is a *Pareto weighting calculation*.

$$\min F(x) = \sum_{m=1}^M w_m f_m(x)$$

$$s.t. \begin{cases} g_j(x) \geq 0 & j = 1, 2, \dots, J \\ h_k(x) = 0 & k = 1, 2, \dots, K \\ x_i^{(L)} \leq x_i \leq x_i^{(U)} & i = 1, 2, \dots, n \end{cases}$$



(a) Original model



(b) Optimized model

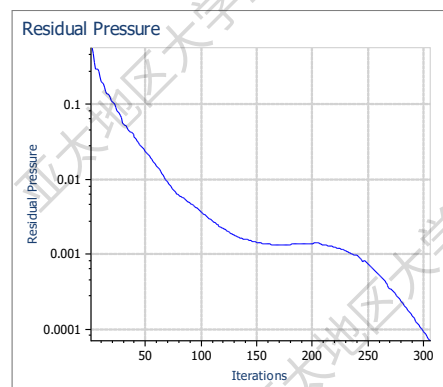
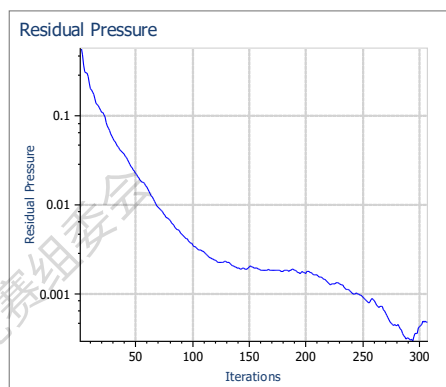


Figure 14 High-speed rail optimization before and after comparison chart.

4.4 Noise model of high-speed train

4.4.1 Status of foreign high-speed train noise research

In recent years, environmental noise pollution caused by traffic noise continues to increase and has become a serious problem in many countries, the following section will introduce respectively The current status of foreign research on high-speed train (railroad) noise.

Researchers in Germany believe that HSR operation noise has a greater impact on the environment along the line than the vibration caused by HSR, when the maximum speed reaches 300km/h or more , noise will increase significantly⁴. See Table 2, 3, 4 for details.

Table 2 The relationship between high-speed train speed and noise level.

Speed/km · h ⁻¹	Noise level/dB(A)
200	82
250	85
300	89
400	102

Table 3 Japan Railway Environmental Noise Standards.

Region	Shinkansen Railway (high-speed railroad) (environmental standards)	New construction of general railroad (guidance)	
	L_{Amax} /dB(6:00-24:00)	Daytime	Nighttime
Residential	70	60	55
Commercial and industrial areas	75	60	55

Table 4 France and Italy railroad noise standard limit value L_{pAeq} /dB.

Country	Daytime	Night
France (high speed)	60	55
France (new general speed)	63	58
France (highway conversion)	65	60
France (conversion of general speed)	68	63
Italy (new highway)	65	55
Italy (existing, reconstruction, new general speed)(0~100m)	70	60
Italy (existing, reconstruction, new general speed)(100~250m)	65	55

⁴High-speed rail model is ICE-V. The measurement point is 25m from the centerline of the track.

4.4.2 Bow wave and Mach cone

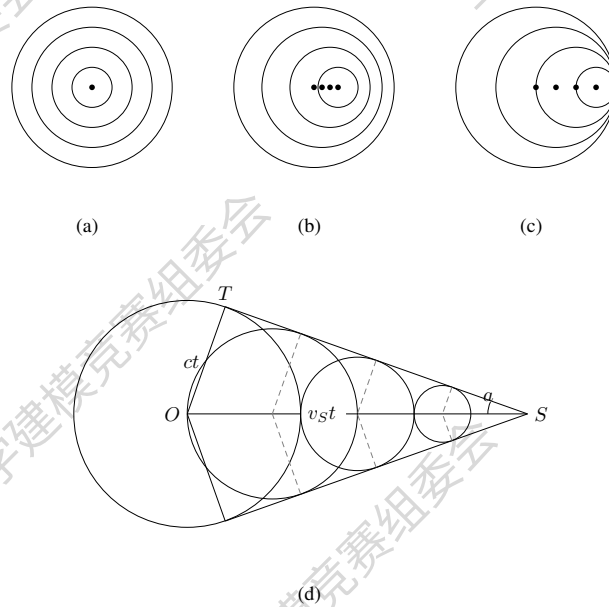


Figure 15 Wavefront of the moving wave source.

The Figure 15a the wave source is stationary, wavefront is concentric; In the figure 15b the wave source is in motion, but its speed is less than the wave speed, the center of the wavefront is misaligned, creating a *Doppler effect*; In the Figure 15c the velocity of the wave source tends to the wave speed and all wavefronts are tangent at one point, frequency $\nu' \rightarrow \infty$; In the Figure 15d the speed of the wave source exceeds the wave speed, the envelope of the wavefront is conical in shape and is called *Mach cone*. Since in this case the propagation of the wave does not exceed the moving object itself, mach cone surface is the leading edge of the wave, no disturbance waves outside of it. This form of fluctuation is called *Bow wave*. Such that the half-top angle of the *Mach cone* is α , As can be seen from the figure:

$$\sin \alpha = \frac{c}{v_s}$$

The dimensionless parameter $\frac{v_s}{c}$ is called the *Mach number*, it is a very useful parameter in aerodynamics. Air under high velocity impact can be considered as being under an adiabatic process, so the pressure change will trigger a change in air temperature.⁵ In the humid air, In humid air, the temperature of the thinnest part of the shockwave air (the wall portion) drops below the dew point, causing the moisture in the air to condense quickly and present a visible cone of fog⁶. And present a visible cone of fog.

Although the speed of high-speed train operation is much less than the speed of sound, the wave surface intensifies the buildup in the interval of the forward direction of the high-speed train (Similar to figure 15b), it makes the air friction increase, and the noise will increase.

⁵The speed of sound in air is about 340m/s at 1 standard atmosphere and 15°C.

⁶In the temperature range where the adiabatic index and the gas constant change negligibly, the speed of sound is proportional to the temperature open root sign.

4.4.3 High-speed rail external Aerodynamic Noise model

The turbulent kinetic energy turbulent dissipation rate at each node in the steady-state flow field is extracted from the air resistance model in turbulence above, and the acoustic power at each node is determined by the expression^[3]:

$$W_A = \alpha \rho \left(\frac{u^3}{l} \right) \frac{u^5}{c_0^5} \quad (13)$$

where α is a constant; l is the turbulence scale; u is the turbulence velocity; and c_0 is the speed of sound.

Then, the large eddy model (see 4.2.1) is used to simulate the transient flow field in combination with the steady-state flow field. The pressure distribution on the surface of the high speed rail is obtained, and finally, the prediction of the aerodynamic noise of the external flow field of the high speed rail is done by using *Lighthill-Curle acoustic analogy theory*, and the expression^[4] is:

$$\rho'(\mathbf{x}, t) = -\frac{1}{4\pi c_0^2} \frac{\partial}{\partial x'_i} \int_S \frac{n_j P_{ij} \left(\mathbf{y}, t - \frac{R}{c_0} \right)}{R} d\mathbf{y} \quad (14)$$

$$P(\mathbf{x}, t) - P_0 = c_0^2 \rho'(\mathbf{x}, t) \quad (15)$$

where $\mathbf{x}$ is the sound measurement point vector; t is the vector; $\mathbf{y}$ is the sound source point vector; n_j is the directional cosine of the vertical high iron outer wall outward (pointing to the outflow field); P_{ij} is the pulsating pressure of air on the outer surface of the body; $\mathbf{R} = \mathbf{x} - \mathbf{y}_j$; P and P_0 are the sound pressure and the reference sound pressure ($P_0 = 2 \times 10^{-5}$ Pa).

4.4.4 Aerodynamic noise Simulation

To better reflect the relationship between high-speed train model and external noise, we reset the wind tunnel speed to 85m/s (306km/s), which is also processed to correlate with the data in Table 2 for the train model. And we follow the shape of Figure 10 to produce a theoretical conical type of high speed rail model as shown in Figure 16 (for the calculations below).

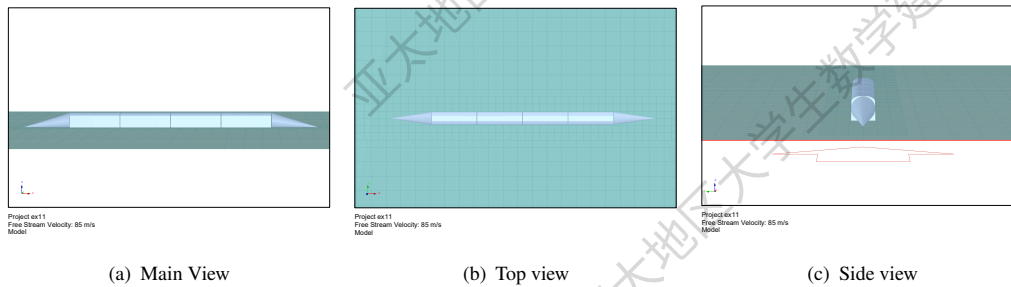


Figure 16 High-speed rail model based on theoretical conical type.

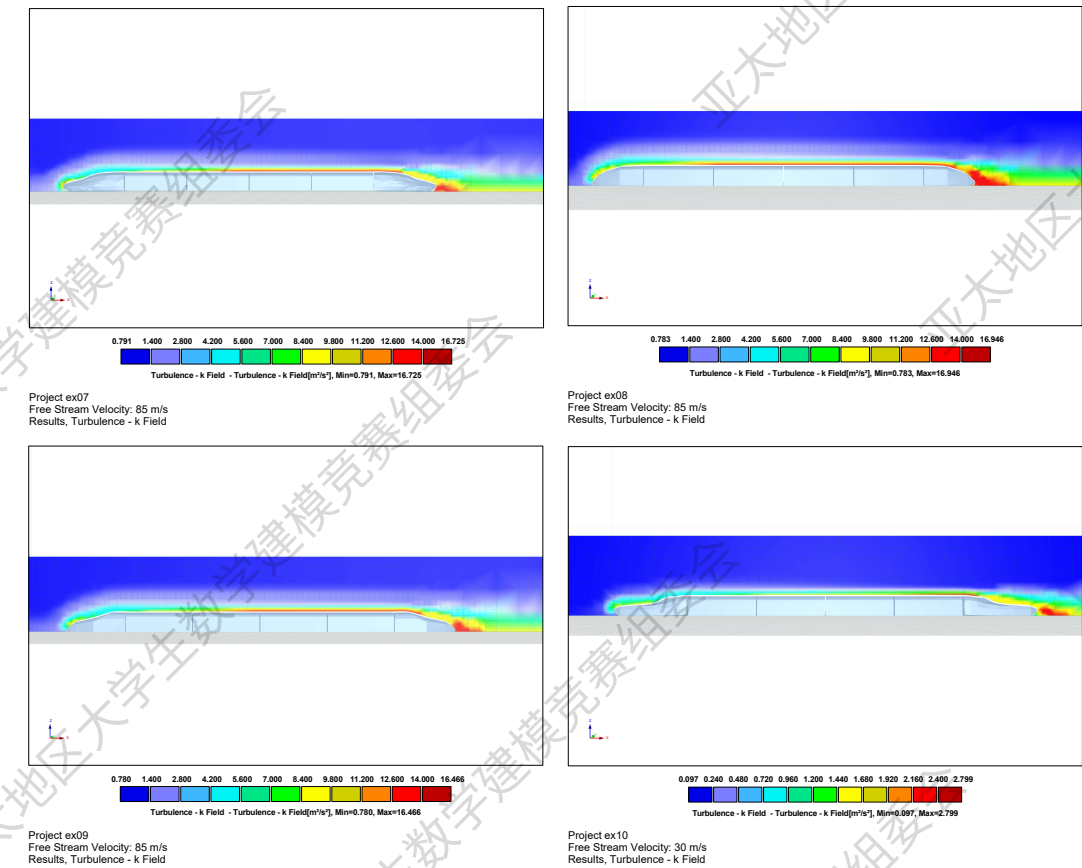


Figure 17 Turbulence - k Field - Turbulence - k Field (Images)[m^2/s^2].

Table 5 Turbulence - k Field - Turbulence - k Field (Values)[m^2/s^2].

	TP1	TP2	TP3	TP4
min	0.791	0.783	0.780	0.097
max	16.725	16.946	16.466	2.799

Compare Figure 17 and Table 5. It can be found that the strong value region of TP4 high speed rail is the smallest, and not only the minimum value is the best, but also the maximum value is much more desirable than the other three models. Therefore, **TP4 is the best shape of high speed rail that generates the least noise**. TP4 is designed to comply with the relevant regulations in Japan to a certain extent.

In summary, it is necessary to analyze the TP1 and TP4 trains more thoroughly and combine their advantages to design a new type of high-speed train that takes into account both speed and noise reduction.

4.5 Optimized model for both Resistance and Noise

4.5.1 Introduction to Multi-Objective Particle Swarm Algorithm

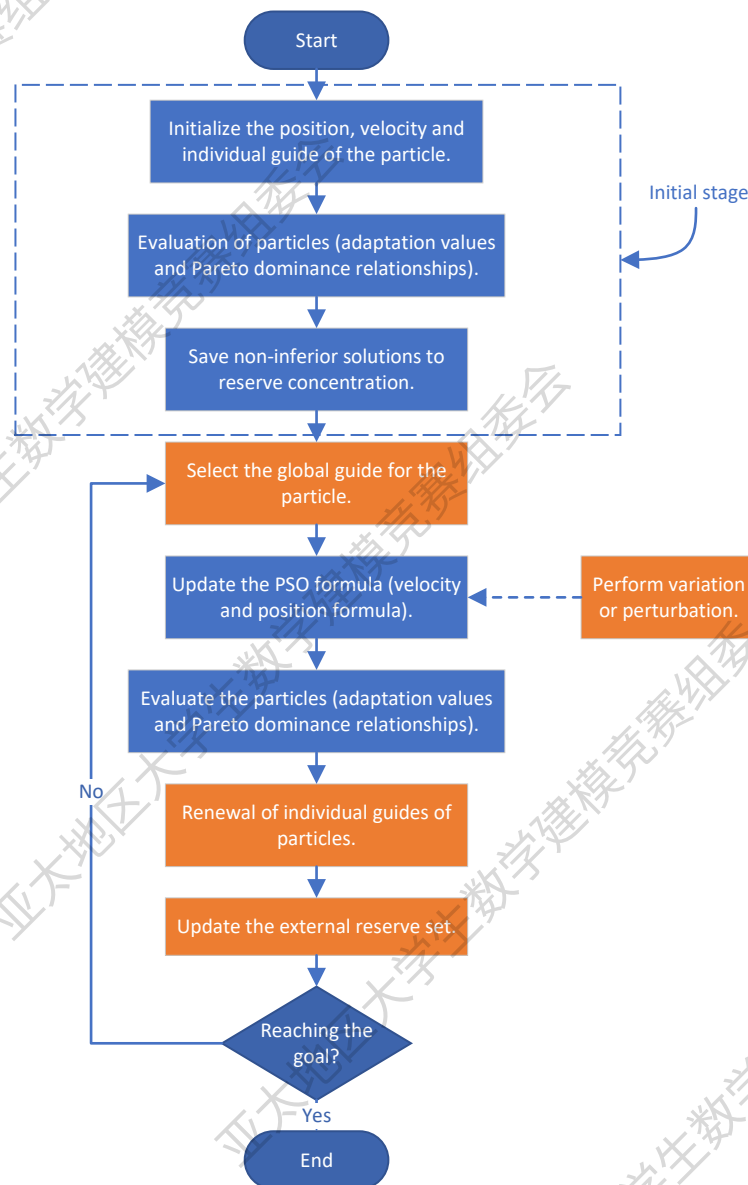


Figure 18 The basic workflow of the MOPSO algorithm.

The expected return and variance (or standard deviation) of all Pareto solutions are expressed on a two-dimensional plane, giving a curve called the Pareto Frontier, where the points above the Pareto Frontier are unavailable and the points below the Pareto Frontier, corresponding to the portfolio, are always available.

In order to obtain a non-inferior solution set that covers the entire search space, is uniformly distributed and close to the true Pareto front with as few computational resources as possible is the goal of the MOPSO algorithm. In this paper, this algorithm will be adopted to adjust the train head model that takes into account both speed and noise.

4.5.2 Numerical solution of the model

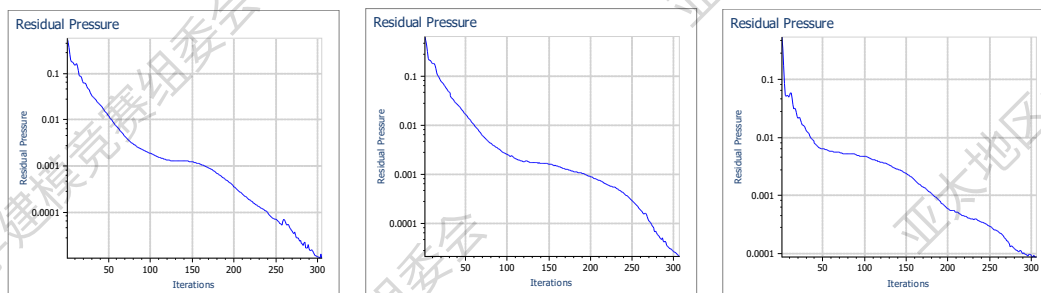


Figure 19 Numerical simulation of TP1 (left) and TP2 (middle) high speed rail with carriages. And numerical simulation of an optimal high-speed train model (right) that balances speed and noise

From the above analysis, TP1 has less air resistance property and TP4 has less noise property, so we made *Figure 17 1 and 4* the relationship between Residual Pressure and iterations as shown in *Figure 19 1 and 2*. We find that the first and last parts of the TP1 curve converge faster, and the middle part of the TP4 curve is flatter, and the designed high speed rail model inherits these two properties. The following are the relevant wind tunnel simulation values for the model, with specific size reference *Figure 23*. Combine *Figure 16* get $\Delta_{TP1} = 0.0105$, $\Delta_{TP4} = 0.0031$, $\Delta_{TP_{best}} = 0.0029$.

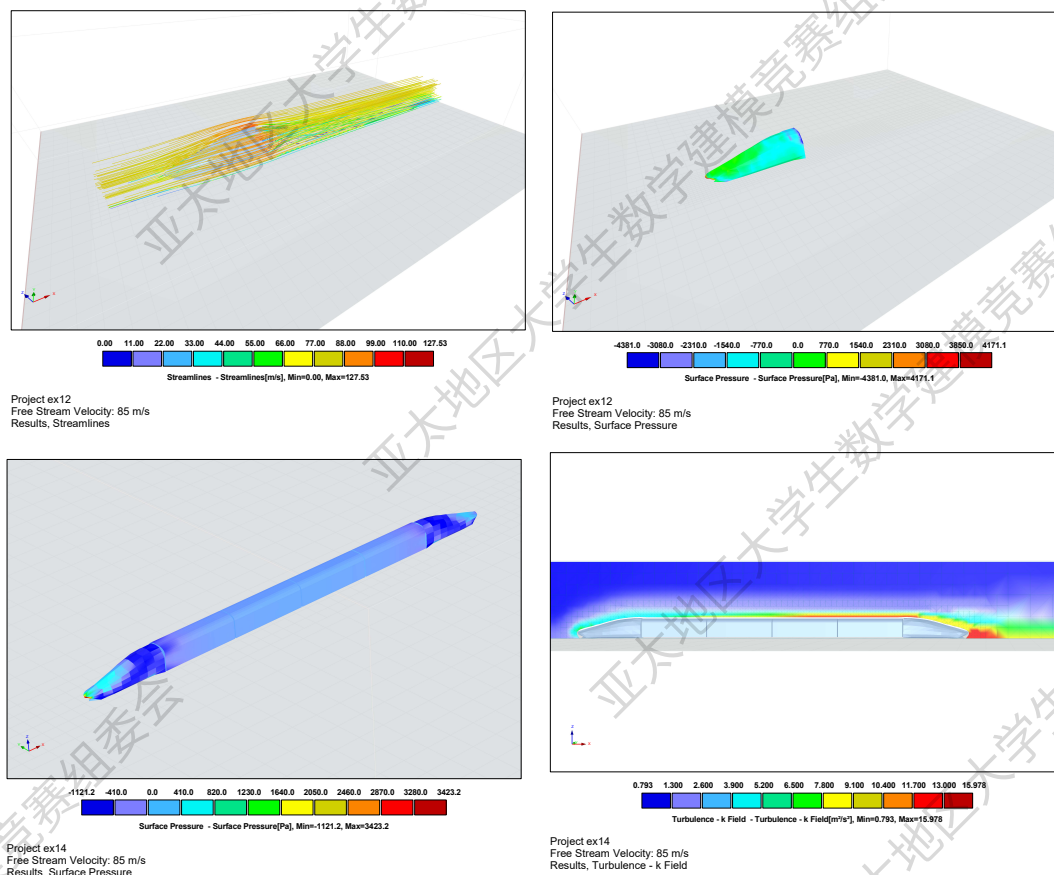


Figure 20 Relevant test data of the integrated and optimized high speed rail model in the wind tunnel.

4.5.3 Internal adjustment of the integrated optimization train model

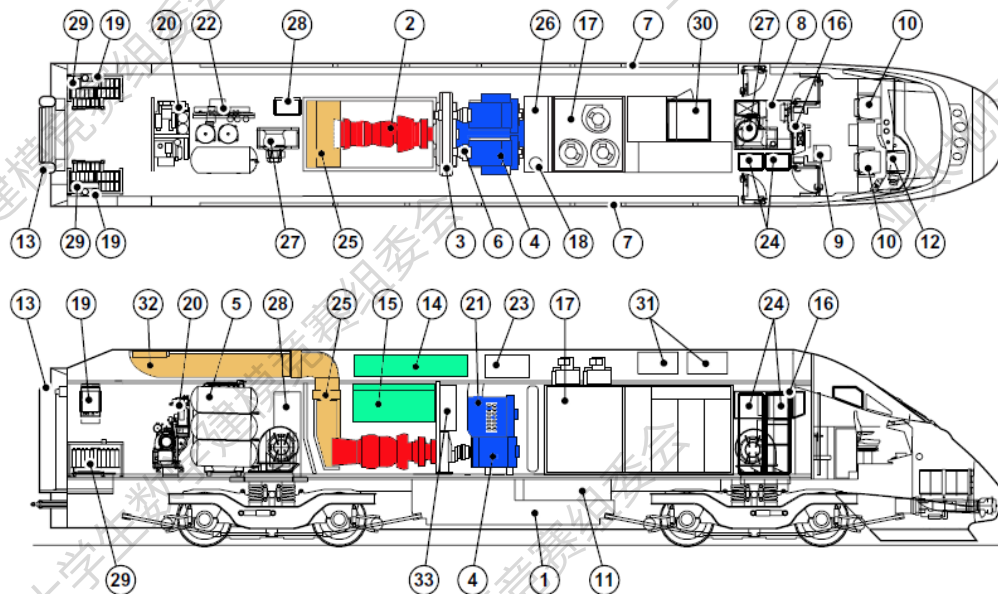


Figure 21 Schematic diagram of the general train interior structure.

Figure 21 is a schematic diagram of the general train interior equipment. For the comprehensive optimization model, we should consider not only its external aerodynamic design, but also the adjustment of the corresponding equipment inside the train.

Comparing the models before and after optimization, it can be found that the nose cone height decreases, the nose cone length increases, and the nose cone becomes smoother; the driver's chamber height is adjusted downward, and the front half of the streamline shape becomes narrower and the width of the rear half increases. So we have to adjust the device numbers accordingly: 8, 9, 10, 12, 16, 24, 27 are rearranged; other device numbers of the corresponding new models are designed.

V. Evaluation of the models

5.1 Advantages of the model

1. The established model is closely related to the actual, general and generalization is strong;
2. The consideration of the model is very comprehensive, the simulation results are reasonable and the results are accurate;
3. The use of wind tunnel experiments, can accurately control the experimental conditions, high precision experiments;

5.2 Disadvantages of the model

1. There is a certain amount of error;
2. For the model can not do exactly the same;
3. There is still a lot of room for improvement.

VI. Optimization of the model

Model for Problem 1

The force analysis can be considered as a three-dimensional case, no longer limited to two-dimensional simple mechanical analysis, and can spend more time in the modeling.

Model for Problem 2

The process of Pareto search can be mentioned over the algorithmic implementation, as well as a more detailed theoretical portrayal of the Pareto frontier can be attempted.

Model for Problem 3

More parameters can be compared, making the results more convincing than mentioning min and max and simple wind tunnel simulations to make the case.

Model for Problem 4

The algorithm of multi-objective particle swarm needs to set the initial number of particle swarms according to the width and breadth of the optimization search, and the particle swarm algorithm is easy to fall into the local optimal solution.

VII. Future work

Future research should more carefully consider the potential impact of high-speed train on environmental noise, such as when it passes through caves or forests. And how to design a new type of environmentally friendly and fast high-speed train needs to be studied in more depth and detail.

For the optimization methods described in the paper, the reader can also use excellent modern optimization algorithms such as Simulated Annealing Algorithm or Heuristic Algorithm.

In this paper, we analyze the optimization process of high-speed trains and hope that it will be useful for scholars who are researching in this field. We will continue to pay attention to the research results in this field in the future, and we hope that more and more scholars will participate in it and contribute to the cause they love.

VIII. References

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- [6] Sun Zhenxu, Yao Yongfang, Guo Dilong, Yang Guowei, Yao Shuanbao, Zhang Ye, Chen Dawei, Li Guibo, Shang Keming, Jia Ling. RESEARCH PROGRESS IN AERODYNAMIC OPTIMIZATION OF HIGH-SPEED TRAINS 1. Chinese Journal of Theoretical and Applied Mechanics[J], 2021, 5311: 51-74 DOI:10.6052/0459-1879-20-205.

IX. Appendix

Listing 1: The MATALB Source code of *Table 1*.

```
present00=importdata('ex00_.txt');
present01=importdata('ex01_.txt');
present02=importdata('ex02_.txt');
present03=importdata('ex03_.txt');
present04=importdata('ex04_.txt');

a00=present00(1:300,3:3);
a01=present01(1:300,3:3);
a02=present02(1:300,3:3);
a03=present03(1:300,3:3);
a04=present04(1:300,3:3);

x=present00(1:300,1:1);

deltaY1 = (a01-a00).^2;
deltaY2 = (a02-a00).^2;
deltaY3 = (a03-a00).^2;
deltaY4 = (a04-a00).^2;

k1=sum(deltaY1);
k2=sum(deltaY2);
k3=sum(deltaY3);
k4=sum(deltaY4);
```

Listing 2: The Source code of *Figure 1 (a)*.

```
\begin{tikzpicture}
\draw[->](0,0.1)arc(165:120:5 and 3);
\draw[->](0,-0.3)arc(165:120:5 and 3);
\draw[->](0,-0.5)arc(165:120:5 and 3);
\draw[->](0,-0.7)arc(165:120:5 and 3);
\draw[->](0,-0.1)arc(165:120:5 and 3);
\draw[->](0,0.3)arc(165:120:5 and 3);
\draw[rotate around={93:(1.8,1.42)}](1.8,1.3) ellipse(0.5 and 0.3);
\draw[rotate around={-45:(0.4,-0.2)}](0,0) ellipse(0.3 and 0.15);
\end{tikzpicture}
```

Listing 3: The Source code of *Figure 1 (b)*.

```
\begin{tikzpicture}
\draw(0,0)arc(80:20:5 and 3);
\draw(0,1)arc(80:20:5 and 3);
\draw(0,-3)--(4,-3);
\draw(0.5,-3)--(0.5,-0.1);
\draw(2.8,-0.9)--(2.8,-3);
```

```

\draw(0.6,0.41) ellipse(0.2 and 0.49);
\draw(0.4,0.45) ellipse(0.2 and 0.48);
\draw[rotate around=-45:(3.2,-0.75)](3.2,-0.75) ellipse(0.15 and 0.38);
\draw(2.5,-0.75)arc(149:100:0.8 and 1.4);
\node[left]at(0.5,-2){$h_{1}$};
\node[left]at(2.8,-2){$h_{2}$};
\node[above]at(0.5,1){$a_{1}b_{1}$};
\draw[->](-0.5,0.45)--(0.2,0.45);
\draw[->](3.8,-1.4)--(3.3,-0.9);
\node[below]at(3.8,-1.4){$p_{2}S_{2}$};
\node[above]at(-0.2,0.45){$p_{1}S_{1}$};
\node[above]at(3.2,-0.2){$a_{2}$};
\node[above]at(3.6,-0.6){$b_{2}$};
\end{tikzpicture}

```

Listing 4: The Source code of Figure 2 (a).

```

\begin{tikzpicture}
\draw(2,2) ellipse(1 and 3);
\draw(2,2)ellipse(0.8 and 2.5);
\draw(10,2) ellipse(1 and 3);
\draw(10,2) ellipse(0.8 and 2.5);
\draw(2,-1)--(10,-1);
\draw(2,5)--(10,5);
\draw(2,-0.5)--(10,-0.5);
\draw(2,4.5)--(10,4.5);
\node[above]at (2,5){$a$};
\node[above]at (10,5){$b$};
\draw(2,-1.1)--(2,-2);
\draw(10,-1.1)--(10,-2);
\draw[->](6,-1.5)--(2,-1.5);
\draw[->](6.3,-1.5)--(10,-1.5);
\node[right]at(6,-1.5){$l$};
\draw[->](7,5.2)--(5,5.2);
\draw(1.9,5)--(0.1,5);
\draw(10,4.5)--(0.9,4.5);
\draw[->](5,4.3)--(7,4.3);
\draw(0.3,2)--(1.8,2);
\draw[->](2.1,2)--(4,2);
\draw(4.2,2)--(8,2);
\draw(8.2,2)--(10,2);
\draw(10.2,2)--(12,2);
\draw[->](0.5,3.8)--(0.5,5);
\node[below]at(0.5,3.8){$r+dr$};
\draw[->](0.5,3.3)--(0.5,2);
\draw[->](1,3.3)--(1,4.5);
\node[below]at(1,3.3){$r$};
\draw[->](1,3)--(1,2);
\node[below]at(10.5,2){$p_{b}$};

```

```

\mode[above]at(6,5.2){$f_{r+dr}$};
\mode[below]at(6,4.3){$f_{r}$};
\mode[below]at(0.5,2){$P_{a}$};
\mode[above]at(4,2){$v$};
\end{tikzpicture}

```

Listing 5: The Source code of Figure 2 (b).

```

\begin{tikzpicture}
\draw(0,0)--(6,0);
\draw(0,4)--(6,4);
\draw(2,0)--(2,4);
\draw[->](2,2)--(6,2);
\draw[->](2,0.5)--(4.5,0.5);
\draw[->](2,1)--(5.4,1);
\draw[->](2,1.5)--(5.9,1.5);
\draw[->](2,2.5)--(5.9,2.5);
\draw[->](2,3)--(5.4,3);
\draw[->](2,3.5)--(4.5,3.5);
\draw(2,0) arc( -90:90: 4 and 2);
\draw(6.5,4)--(7,4);
\draw(6.5,2)--(7,2);
\draw[->](6.8,3.2)--(6.8,4);
\mode[below]at(6.8,3.2){$R$};
\draw[->](6.8,2.8)--(6.8,2);
\draw[loosely dashdotted](0,2)--(6,2);
\end{tikzpicture}

```

Listing 6: The Source code of Figure 15 (d).

```

\begin{tikzpicture}
\draw(0,0)circle(2);
\draw(1.5,0)circle(1.5);
\draw(3,0)circle(1);
\draw(4.5,0)circle(0.5);
\draw(0,0)--(2,0);
\draw(0.67,1.89)--(6,0);
\draw(2.8,0)--(6,0);
\draw(0.67,-1.89)--(6,0);
\draw(0,0)--(0.67,1.89);
\draw(0,0)--(0.67,-1.89);
\draw[densely dashed, gray](2,1.414)--(1.5,0);
\draw[densely dashed, gray](2,-1.414)--(1.5,0);
\draw[densely dashed, gray](3.33,0.943)--(3,0);
\draw[densely dashed, gray](3.33,-0.943)--(3,0);
\draw[densely dashed, gray](4.667,0.471)--(4.5,0);
\draw[densely dashed, gray](4.667,-0.471)--(4.5,0);
\mode[right]at(6,0){$S$};
\mode[right]at(2,0){$v_{S}t$};

```



```

\draw[black] (5.3,0) arc (180:155 :0.5) node[above, midway] {$a$};
\end{tikzpicture}

```

Table 6 Simulation values of four typical high speed trains.

No.	TP1	TP2	TP3	TP4
1	0.36000	0.33897	0.28682	0.38613
2	0.61573	0.65582	0.61214	0.63859
3	0.24439	0.24061	0.25190	0.26011
4	0.11960	0.11287	0.10850	0.12508
5	0.07326	0.07139	0.06738	0.08075
6	0.07238	0.07194	0.07130	0.08020
7	0.07843	0.07752	0.08235	0.08330
8	0.08850	0.08329	0.09854	0.08680
9	0.09990	0.09240	0.11668	0.92158
10	0.11683	0.11307	0.14264	0.10541
...	...	...	...	...
39	0.01414	0.01301	0.01213	0.01407
40	0.01346	0.01254	0.01189	0.01368
41	0.01318	0.01220	0.01181	0.01334
...	...	...	...	...
71	0.00802	0.00763	0.00873	0.00845
72	0.00794	0.00757	0.00866	0.00836
73	0.00784	0.00752	0.00861	0.00827
74	0.00776	0.00746	0.00869	0.00819
75	0.00770	0.00741	0.00872	0.00809
76	0.00769	0.00733	0.00880	0.00799
77	0.00760	0.00726	0.00891	0.00788
78	0.00755	0.00722	0.00902	0.00778
79	0.00745	0.00722	0.00911	0.00765
80	0.00735	0.00725	0.00918	0.00753

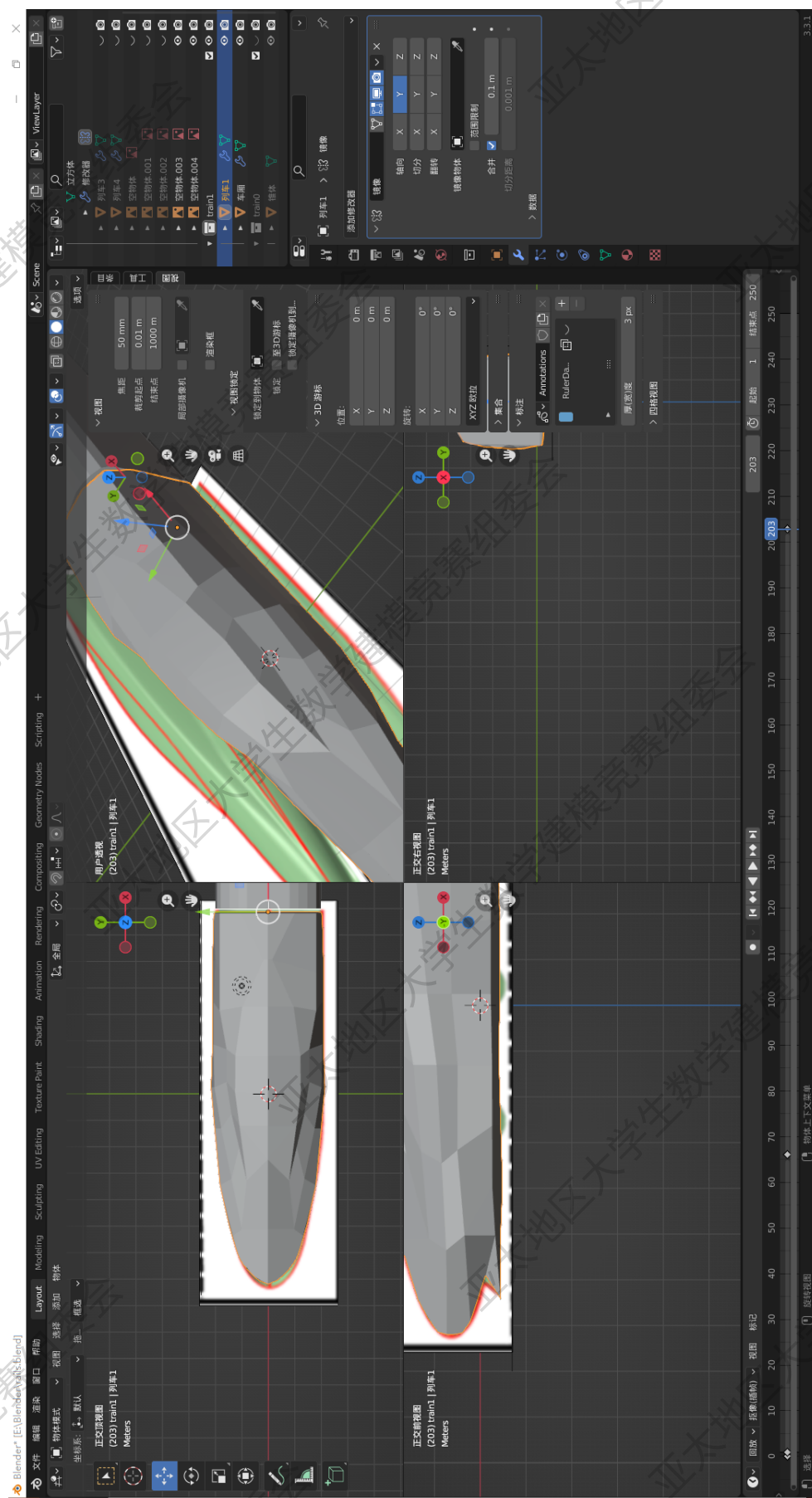


Figure 22 Production process.

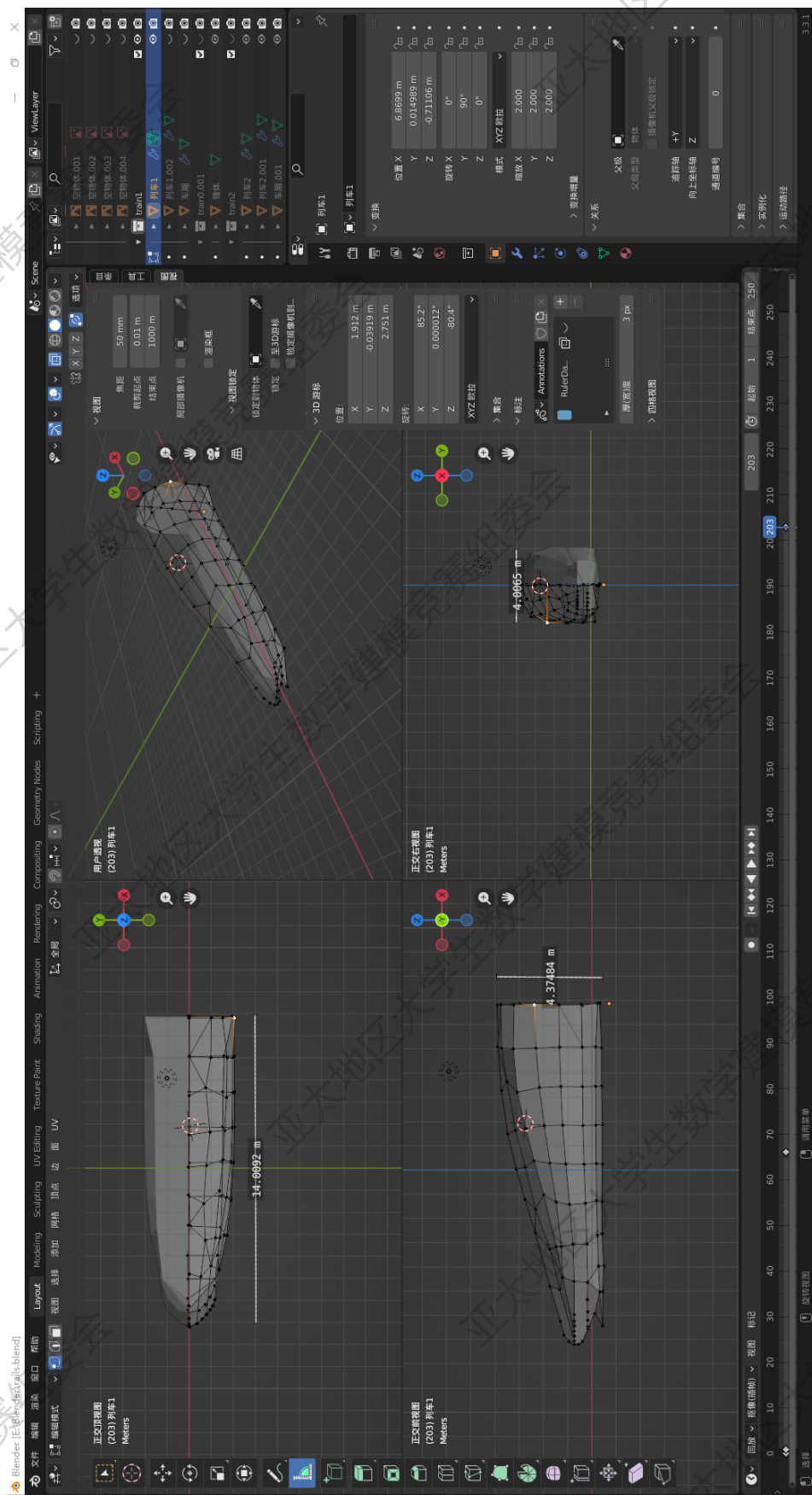


Figure 23 The sketch of the shape of the high speed rail is the corresponding structural parameters.